

Durand Raphaël

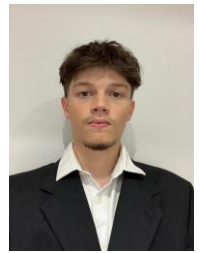
Born on 31 May 2003.

10 Av Félix faure, 75015 Paris

raphael.durand@edu.ece.fr

+33 6 38 94 98 01

Driver's License



Currently completing a final-year internship in « Deep learning models for enhancing medical imaging quality», I am seeking a **PhD in computer science engineering** starting in **September 2026** in the field of **Deep learning** applied to **medical imaging**.

PROFESSIONAL EXPERIENCE

Deep Learning Internship in Medical Imaging (MRI) – Chipiron

February to August 2026, 6-month final-year internship:

Mission: Image-assisted contrast recovery from highly accelerated post-contrast low-field MRI acquisitions.

Tasks:

- **Literature Review:** Studying recent methods in contrast synthesis, image-to-image translation, multi-contrast reconstruction, and accelerated MRI.
- **Development and Integration:** Designing and implementing deep learning architectures (U-Net, DeepCascade, DiT) for the reconstruction of contrast-enhanced MRI images.
- **Data Processing and Evaluation:** Developing preprocessing and evaluation pipelines using both quantitative and qualitative assessment metrics.

Results:

Application of the developed models and methods to improve diagnostic quality in low-field MRI company.



Deep Learning / Image Processing Internship (MRI) – IMB

April to August 2025, 4-month internship:

Mission: Comparison of Deep Learning models for MRI image reconstruction from undersampled data.

Tasks:

- Implemented three end-to-end reconstruction models (Unet, Automap, DeepCascade) using PyTorch and trained them on a public dataset.
- Performed fine-tuning and optimization through parameter and hyperparameter adjustment.
- Developed mathematical formulations of the models to demonstrate their stability.

Results: Produced results that served as a scientific foundation for two complementary PhD theses.



ACADEMIC PROJECTS

Clinical Decision Support Software in Podiatry – Final Year Project

September 2025 – January 2026:

Team development of a podiatric diagnostic support tool based on field studies with practitioners, aimed at improving the speed and reliability of clinical assessments. The tool centralizes patient data, suggests diagnostic leads, and automatically generates reports, with validation by a partner podiatrist.

Skin Lesion Pre-Diagnosis Website – Data Science Project (OA)

November – December 2025

Development of a software application using a neural network fine-tuned from a foundation model, designed to support the diagnosis of skin lesions by classifying moles as benign or suspicious.

Medical Appointment Scheduling Application – Java Project (ECE)

May – June 2024:

Team development of a Java-based medical appointment management application using an MVC architecture and a JDBC database. Responsible for the graphical interface and front/back-end integration. This project strengthened software design and data management skills.

EDUCATION

ECE Paris, Engineering school

Major «Health & Technology»

From September to December 2023:

International semester in New York (Baruch College)

LANGUAGES

English (B2 : 830 TOEIC)

German (B1)

SKILLS

- **Programming languages:** python (pytorch, scikit-leran numpy, matplotlib, pandas), C, MATLAB.
- **Tools :** Git, Linux, VSC
- **Project Managment :** Project report writing, oral presentations

CERTIFICATIONS

- MOOC Project Management Classic Track, Centrale Lille
- MOOC Nanotechnology, University Paris-Saclay
- TechAway Gen IA, Microsoft
- DataScientist - Python